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Commentary

The Golden State Killer investigation and the nascent field of forensic genealogy

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ABSTRACT

The likely genetic analysis steps taken to identify suspect Joseph DeAngelo in the recently resolved Golden State Killer investigation are discussed. The consequences for the forensic genetics community of introducing much more detailed SNP analysis regimes, as used by the Golden State Killer investigators, are reviewed along with some of the limitations in accuracy and sensitivity that may be involved in such approaches.

As a long-standing advocate of the application of genomics to forensic DNA analysis, I followed news reports of the successful detection of the Golden State Killer with increasing interest. Much of the press coverage centered on the ethical issues raised by certain actions taken by the investigators to identify Joseph DeAngelo, namely: surreptitious sample collection; accessing the community-sourced GEDmatch database to detect a criminal suspect – a "citizen science" website originally established to help reunite kinship members or adoptees with their biological parents; and passing off the variant data obtained using DNA from a preserved rape kit from the 80's, as a personal genome report. The companion commentary by Denise Syndercombe Court discusses these and other ethical aspects to the case. Here I will review the genetics involved (as far as this is possible to discern from often scant or inaccurate reportage), and the consequences of the analyses made by the Golden State Killer investigation, herein GSKI, for forensic DNA analysis and investigative practice in the future.

While the description of the genomic analysis steps taken to find DeAngelo appear to be the latest groundbreaking technology, investigators have been considering the use of freely accessible personal genetic testing data for some time. This is partly a response to their access being barred to customer data held by the four biggest US direct-to-consumer (DTC) genetic testing vendors: AncestryDNA; 23andMe; Family Tree DNA; MyHeritage (in order of user base). There has been a degree of success from extending the concept of familial searching beyond CODIS data and its confinement to close relative matches. For example, in 2015 Bryan Miller was identified as prime suspect in the Phoenix Canal Killer case by genealogist Colleen Fitzpatrick who uploaded a Y-Filer profile to a number of public Y-STR databases (probably Ysearch, the Sorenson Molecular Genealogy Foundation and AncestryDNA, although the latter has ceased offering Y-STR tests) [1]. Interestingly, Fitzpatrick had volunteered her genealogical services to the forensics community after giving a talk at the 25th ISHI conference in Phoenix the year before [2]. Y-STR analysis started much of the initial developments in genetic genealogy and the tools hosted by GEDmatch evolved from a 'Rogers' surname-matching group run by its co-founder Curtis Rogers [3]. GSKI first tried limited Y-STR matching in 2017 (one of the genotypes was described as atypical in European Y lineages), but this misidentified a 73-year old man in Oregon. A recent report indicates that GSKI used the open-access Ysearch database (now closed) and issued a subpoena to Family Tree DNA to reveal the details of a match of 12 Y-STRs [4] - instigating this failed investigative lead. As early as 2014, investigators were using a Y-STR profile to attempt the identification of DNA from the murder of Angie Dodge, but analyses were similarly restricted by a lack of genetic data and an innocent person was misidentified by the adventitious hit obtained in this case [5]. It might be the case that a study in 2013 by Gymrek et al. [6] indicating "triangulation" methods could identify otherwise anonymous DNA data in open genealogical databases (combining Y-STR genotypes with other metadata) led to over-confidence in adopting the same approach for criminal investigations. However, the major advances in genetic genealogy in unknown parentage and missing persons cases have come from use of large-scale autosomal marker sets, not Y data [7].

Genealogists have identified missing persons with increasing precision, as an additional form of analysis to the established help given to those wishing to discover their distant relatives using the proprietary company databases and GEDmatch. This is the inevitable outcome of both the large number of individuals accumulated by vendor- or community-databases, and the density of the genetic marker data all four vendors provide. At the time of writing, AncestryDNA has 10 million DNA customers and 23andMe 5 million, while the other vendors and GEDmatch have about 1 million each. All vendors use customized versions of the Illumina *OmniExpress* Array or Illumina *Global Screening* Array which both comprise over 650,000 loci. A key study in 2012 by

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Henn, et al. [8] showed that 2nd to 9th cousinships were detectable using identity-by-descent (IBD) measurements of a 550 K SNP intersect from the 650 K SNP data of the HGDP-CEPH panel and the 580 K SNPs genotyped by 23andMe at that time. Therefore, the tools to detect close and distant relatives using high-density SNP datasets have been in place for some time. These consist of: i. IBD analysis that can detect very distant cousin pairs and thus provides a depth of information that can range from manageable numbers of related individuals sharing great-grandparents, up to much more extensive but distant kinships; ii. high density SNP data from the DTC vendors to allow customers to query the vendor's proprietary databases or share their data in GEDmatch to broaden a search out to the genealogy community (GEDmatch has a *Genesis* database allowing comparisons between data from different SNP arrays); and iii. growing availability of extensive online genealogy databases (e.g. censuses; birth, marriage and death indices; digitized newspapers; etc.). As a whole, these factors allow detailed family tree reconstruction, which not only routinely handles centimorgan values, but compiles a wide range of other details, including birthdates, places of residence and kinship member's personal characteristics (see [9] for the blog details of three authoritative practitioners). It is notable that GEDmatch have changed their terms and conditions to directly highlight the possibility of investigative use of community data held on the website and now allows users to withdraw their data; but at the same time, they have made it easier for investigators to upload probative DNA data without the need to use an alias. These steps are to be commended, as transparency is a crucial factor in preserving user's trust and it is important that those who are not concerned about distant relatives being detected by police are free to continue exploring their personal data and benefiting from the custom tools that have made GEDmatch so effective for this purpose.

Given the developments described above, it might seem surprising that police have only just started to obtain major investigative breakthroughs (and when they do, three separate cases identifying murder victim Marcia King [10], and suspects Joseph DeAngelo and William Earl Talbott [11] are reported within weeks of each other). Yet the obvious missing piece in the jigsaw of genomic analyses made in such cases is how they were able to generate a sufficient amount of SNP data from limited evidential material. In the absence of information from GSKI, a large degree of supposition is needed here to reconstruct the processes likely to have been used to obtain the SNP data eventually uploaded to GEDmatch. Therefore, more detail may eventually come to light that changes this perspective and the purported steps taken might be inaccurate in specific details. Initially, DNA was taken from a duplicate rape kit [12] and treated as single source (no details of the applied extraction processes are available), then analyzed with a SNP array compatible with the DTC vendor databases [4]. If the SNP genotypes generated centimorgan values from "chromosome painting" that revealed a specific kinship, connected in part to the geographic distribution of criminal activity, then investigators could begin to compile supporting details about places of residence, lifestyle and physical characteristics of age- and sex-matched kinship members. This process can provide an effective approach even with very extensive family trees, of which only a handful of members may have had their DNA tested by DTC vendors. The information gathered does not always need to be particularly specific: Bryan Miller was one of tens of thousands of Millers living in Phoenix, but the only one on a list of suspects at the time [1]. Although the principal vendor- and community-databases hold much more data from US customers than from other countries, the application of the genealogical analyses described here have inevitably brought interest from police elsewhere, such as Sweden, where an estimated 40,000 people have taken a genetic ancestry test [13].

Descriptions of large-scale genomic analyses may seem disconcerting when, until recently, forensic DNA analysis has dealt with just a small number of well-established STRs and has not used sequencing beyond the analysis of mtDNA. The jump from 24 A-STRs and 27 Y-STRs to more than 650,000 SNPs is a quantum leap indeed, and most forensic laboratories will not have the bioinformatics infrastructure or analysis skills that such data scales demand. SNPs have been perceived to be markers reserved for small-scale bespoke tests still at an early stage of forensic use, such as ancestry analysis and phenotype prediction. Equally, the utmost care is taken in forensic testing to control contamination; to ensure genetic markers are fully validated; and to confirm the genotypes are specific to one donor, not a mixture of signals from multiple DNA sources. It can also be suggested that investigators have now found a way to circumvent the carefully prepared checks and balances put in place to regulate familial searches of national DNA databases. The important context here is the difference between standardized forensic identification practices and obtaining new genetic information by novel means that might provide vital investigative leads. In routine forensic practice, we all rightly concern ourselves with precision and sensitivity; in other words, it is critical to produce the best possible DNA profile from minimal contact traces and which uniquely identifies an individual. An investigation does not need to deal in such certainties and the police will willingly add the extra detail DNA can provide to the body of information already gathered in a complex criminal enquiry. Investigators can process information on a scale of reliability, usefulness and relevance to the direction of enquiries – akin to the *Admiralty System* for evaluating compiled items of intelligence [14]. I recently dismissed the suggestion that DNA from a licked stamp on a letter sent by the Zodiac Killer [15] would be sufficient to obtain a whole genome sequence or SNP array data with over half a million genotypes. Yet modern sequencing technologies potentially open the door to detailed genomic analysis of minimal DNA and are likely to show such perceptions from an orthodox forensic viewpoint are too pessimistic. We could be witnessing the emergence of genomic genealogy as a completely new field in forensic genetics – one that uses the extended haplotype block, not STRs, as the forensic marker of choice.

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References

[1] Phoenix Canal Killer genealogical analysis: https://eu.azcentral.com/story/news/local/phoenix/2016/11/30/how-forensic-genealogy-led-arrest-phoenix-canal-killer-case-bryan-patrick-miller-dna/94565410/.
[2] 25th ISHI discussions on Y-STR lineage analysis: https://www.ishinews.com/solving-the-phoenix-canal-murders/.
[3] Growth and development of GEDmatch: https://www.theatlantic.com/science/archive/2018/06/gedmatch-police-genealogy-database/561695/.
[4] GSKI misdirection in 2017 using Y-STRs to an unrelated man in Oregon and subsequent success with a high density SNP set: https://www.buzzfeed.com/peteraldhous/family-tree-dna-subpoena-golden-state-killer.
[5] The misidentification of Michael Usry Jr. in the Angie Dodge murder investigation: http://www.postregister.com/articles/chris-tapp-coverage-featured-news-daily-email/2017/07/29/contradictory-dna-results-put.
[6] M. Gymrek, A.L. McGuire, D. Golan, E. Halperin, Y. Erlich, Identifying personal genomes by surname inference, Science 339 (2013) 321–324.
[7] Initial development of genetic genealogy: http://jogg.info/pages/vol8/editorial/moore/moore-history.html.
[8] B.M. Henn, L. Hon, J.M. Macpherson, N. Eriksson, S. Saxonov, I. Pe'er, J.L. Mountain, Cryptic distant relatives are common in both isolated and cosmopolitan genetic samples, PLoS One 7 (2012) e34267.
[9] 'Cruwys news', the blog of genetic genealogist Debbie Kennett: https://cruwys.blogspot.com 'Your Genetic Genealogist' the blog of genetic genealogist CeCe Moore: http://www.yourgeneticgenealogist.com/ 'DNAeXplained', the blog of

genetic genealogist Roberta Estes: https://dna-explained.com.
[10] Identification of murder victim “Buckskin Girl”, Marcia King: http://cases.dnadoeproject.org/mesmerize/buckskin-girl/.
[11] Identification of murder suspect William Earl Talbott: https://www.washingtonpost.com/news/morning-mix/wp/2018/05/21/a-genealogy-website-used-to-crack-another-cold-case-police-say-this-one-a-1987-double-homicide/?utm_term=.2ae49615aa.
[12] Storage and use of a duplicate rape kit in the GSKI: https://abcnews.go.com/US/inside-terrifying-golden-state-killer-crime-spree-key/story?id=54849196.
[13] Swedish police consider accessing Swedish DTC data: https://www.thelocal.se/20180507/swedish-police-mull-using-dna-family-tree-websites.
[14] E. Bruenisholz, A. Ross, S. Prakash, C.P. Roux, M. Morelato, T. Raymond, S. Walsh, T. O’Maley, The Intelligent Use of Forensic Data, Researchgate PDF at: https://www.researchgate.net/publication/286930641_The_Intelligent_Use_of_Forensic_Data.
[15] Genealogical analysis proposed for Zodiac Killer DNA: https://www.independent.co.uk/news/world/americas/zodiac-killer-golden-state-investigation-dna-tests-california-vallejo-sacramento-a8343086.html.